Impact of high-dose folic acid supplementation in pregnancy on biomarkers of folate status and 1-carbon metabolism: An ancillary study of the Folic Acid Clinical Trial (FACT)

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ABSTRACT

Background: Periconceptional folic acid (FA) supplementation is recommended to prevent the occurrence of neural tube defects. Currently, most over-the-counter FA supplements in Canada and the United States contain 1 mg FA and some women are prescribed 5 mg FA/d. High-dose FA is hypothesized to impair 1-carbon metabolism. We aimed to determine folate and 1-carbon metabolism biomarkers in pregnant women exposed to 1 mg or 5 mg FA.

Objectives: This was an ancillary study within the Folic Acid Clinical Trial (FACT), a randomized, double-blinded, placebo-controlled, phase III trial designed to assess the efficacy of high-dose FA to prevent preeclampsia.

Methods: For FACT, women were randomized at 8–16 gestational weeks to receive daily 4.0 mg FA (high dose) or placebo (low dose) plus their usual supplementation ($\leq$1.1 mg). Women were recruited from 3 Canadian FACT centers and provided nonfasting blood samples at 24–26 gestational weeks for measurement of RBC and serum total folate, serum unmetabolized FA (UMFA), tetrahydrofolate (THF), 5-methylTHF, 5-formylTHF, 5,10-methenylTHF, and MeFox (pyrazino-s-triazine derivative of 4α-hydroxy-5-methylTHF, a 5-methylTHF oxidation product); total vitamins B-12 and B-6; and plasma total homocysteine. Group differences were determined using χ^2, Fisher exact, and Wilcoxon rank-sum tests.

Results: Nineteen (38%) women received high-dose FA and 31 (62%) received low-dose FA. The median RBC folate concentration was 2701 (IQR: 2243–3032) nmol/L and did not differ between groups. The high-dose group had higher serum total folate (median: 148.4 nmol/L, IQR: 110.4–181.2; $P = 0.007$), UMFA (median: 4.6 nmol/L, IQR: 2.5–33.8; $P = 0.008$), and 5-methylTHF (median: 126.6 nmol/L, IQR: 98.8–158.6; $P = 0.03$) compared with the low-dose group (median: 122.8 nmol/L, IQR: 99.5–136.0; median: 1.9 nmol/L, IQR: 0.9–4.1; median: 108.6 nmol/L, IQR: 96.4–123.2, respectively). Other biomarkers of 1-carbon metabolism did not differ.

Conclusions: High-dose FA supplementation in early pregnancy increases maternal serum folate but not RBC folate concentrations, suggesting tissue saturation. Higher UMFA concentrations in women receiving high-dose FA supplements suggest that these doses are supraphysiologic but with no evidence of altered 1-carbon metabolism. *Am J Clin Nutr* 2021;113:1361–1371.

Keywords: supplementation, folate, folic acid, folate status, pregnancy

Introduction

Folic acid (FA) intake in the periconceptional period reduces the risk of neural tube defects (NTDs) (1, 2). Mandatory FA fortification of white wheat flour and other enriched cereal products was introduced by Canada and the United States, among others, to reduce the incidence of NTDs (3, 4). Since the implementation of fortification, the prevalence of NTDs has decreased significantly (5, 6), and folate deficiency in the general population is virtually nonexistent (7). Despite the success of FA fortification, not all women of childbearing age achieve an RBC folate status that meets the WHO cutoff of 906 nmol/L associated with maximal NTD risk reduction (8). Approximately 25% of nonpregnant Canadian women who do not consume FA-containing supplements have RBC folate concentrations <906 nmol/L (9). It therefore remains important that women consume FA-containing supplements in the periconceptional period to reduce the risk of an NTD-affected pregnancy.

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Women at low risk for an NTD-affected pregnancy are advised to consume a daily multivitamin containing 400 μg FA in Canada (10) and 400–800 μg FA in the United States (11). For women at high risk for having an NTD-affected pregnancy, many health professional organizations recommend higher FA doses in the range of 4.0–5.0 mg (12–14). At issue is which women stand to benefit from higher doses, given the variable definitions among organizations and jurisdictions for who is at high risk for an NTD-affected pregnancy (11, 12, 15, 16). The US Preventative Services Task Force identifies high-risk women as those who have had a previous NTD-affected pregnancy (11). In Canada, a woman is at high risk if she or her male partner has a personal NTD history or has previously had an NTD-affected pregnancy (12). In the United Kingdom, women with type 1 or 2 diabetes or obesity are also considered at high risk (17, 18), despite an absence of evidence that NTDs associated with these conditions are folate responsive. In Canada, women with type 1 or 2 diabetes are considered at moderate risk and are advised to consume 1 mg FA/d (12). Variable NTD risk definitions across jurisdictions, combined with consumption of FA-fortified foods and FA-containing supplements, can result in women consuming FA at doses exceeding the tolerable upper intake level (UL) of 1 mg/d (19, 20).

Potential adverse effects of FA intake exceeding the UL in pregnancy, if any exist, remain unclear (21). The UL is based on the potential for high-dose FA to exacerbate vitamin B-12–dependent neurodegeneration, an issue that remains controversial (22, 23). Although FA supplementation is effective at increasing RBC and serum total folate concentrations (24), circulating unmetabolized FA (UMFA) is ubiquitous in populations consuming fortified foods, and high UMFA concentrations are commonly observed among pregnant and postpartum women (25–27). It has been hypothesized that UMFA, through its metabolism to dihydrofolate, can impair folate-mediated 1-carbon metabolism by inhibiting key folate-dependent enzymes (28–30). The objective of this study was to compare the effects of high and low doses of FA supplementation on folate status and other biomarkers of folate-mediated 1-carbon metabolism.

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Supplemental Tables 1–4 and Supplemental Figure 1 are available from the "Supplementary data" link in the online posting of the article and from the same link in the online table of contents at https://academic.oup.com/ajcn/.

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Abbreviations used: DFE, dietary folate equivalent; FACT, Folic Acid Clinical Trial; FA, folic acid; LOD, limit of detection; MeFox, pyrazino-s-triazine derivative of 4α-hydroxy-5-methylTHF, a 5-methylTHF oxidation product; MTHFD1, methylenetetrahydrofolate dehydrogenase 1; MTHFR, methylenetetrahydrofolate reductase; MTR, methionine synthase; NTD, neural tube defect; PlGF, placental growth factor; PLP, pyridoxal 5′-phosphate; sEng, soluble endoglin; sFlt-1, soluble fms-like tyrosine kinase 1; SNP, single-nucleotide polymorphism; THF, tetrahydrofolate; UL, upper intake level; UMFA, unmetabolized folic acid; UPLC, ultra performance liquid chromatography.

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Methods

Study design

The Folic Acid Clinical Trial (FACT) was an international multicenter randomized, double-blinded, placebo-controlled, phase III trial conducted to establish the effect of daily high-dose supplementation of FA on the risk of developing preeclampsia among high-risk pregnancies (31, 32). In brief, women were eligible for FACT if they had one or more of the following risk factors for preeclampsia: preexisting hypertension, prepregnancy diabetes (type 1 or 2), preeclampsia in a previous pregnancy, BMI (in kg/m^2) $\geq$35 kg/m^2, and twin pregnancy. Eligible participants were randomly allocated to receive either 4.0 mg/d FA or placebo from randomization (8–16 gestational weeks) until delivery. Participants were allowed to take prenatal vitamins or regular daily multivitamins containing up to 1.1 mg of FA. Thus, participants in FACT consumed either a high-dose FA supplement (4.0–5.1 mg/d) or a low-dose FA supplement ($\leq$1.1 mg/d). FACT included 4 follow-up visits at 24–26 gestational weeks, 34–36 gestational weeks, postdelivery, and 42 d postpartum. Compliance to treatment was >75% in FACT (31). The study described here was nested within FACT and reports on markers of folate status and 1-carbon metabolism in a subset of participants attending the second FACT visit (24–26 gestational weeks). To be included in the current analysis, participants had to be enrolled in FACT and provide informed consent for the ancillary study. No additional eligibility criteria were applied.

Participant recruitment and follow-up

Participants for this study were a convenience sample of women enrolled into FACT at 1 of 3 participating Canadian high-risk obstetric referral centers: The Ottawa Hospital (Ottawa, Ontario), Kingston General Hospital (Kingston, Ontario), and the Moncton Hospital (Moncton, New Brunswick). Women returning for their first FACT follow-up visit at 24–26 completed gestational weeks were approached to participate in the biomarker study (**Figure 1**). Enrollment or withdrawal from this study did not preclude participation in the larger FACT trial. For participants who gave written, informed consent, a nonfasting blood sample was drawn for genetic and biomarker analyses. Recruitment for the current study began in June 2013 and was completed in February 2015; a total of 51 participants consented to blood sample collection and analysis. Participant treatment arm allocation (high-dose or low-dose FA) remained blinded until the FACT trial and all biomarker analyses were complete.

Descriptive characteristics

Baseline participant characteristics, maternal medical history, and pregnancy outcomes were obtained from the FACT data set and included the hospital site (Ottawa/Kingston/Moncton), demographic information (i.e., maternal age, education, employment, ethnic origin), risk factors for obstetrical complications (i.e., chronic hypertension, diabetes, history of preeclampsia, obesity, twin pregnancy), health behaviors (i.e., smoking and alcohol use), and FA supplementation. Other variables included

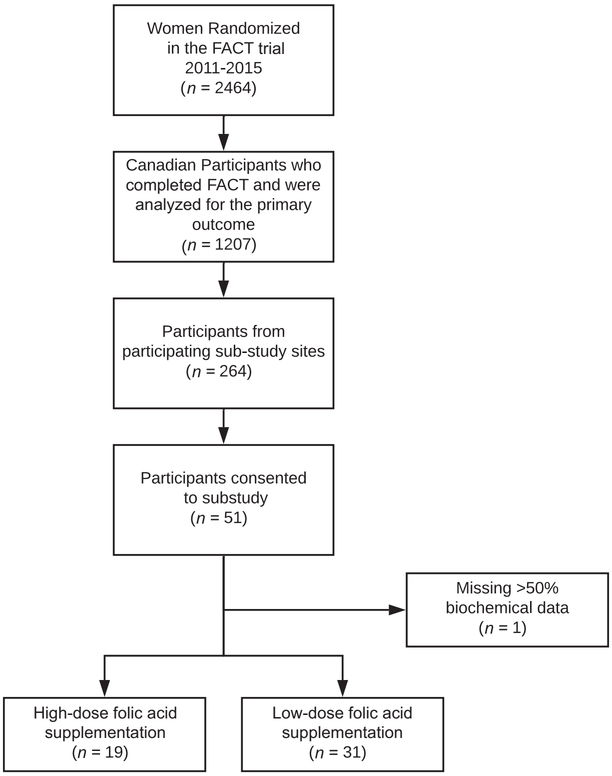

FIGURE 1 Derivation of study participants from the Folic Acid Clinical Trial (FACT) trial. A total of 50 participants from participating Canadian FACT sites in Ottawa, Kingston, and Moncton contributed data to this study. One participant taking low-dose folic acid supplementation was excluded for missing data. In the final cohort, 19 were taking high-dose folic acid supplementation and 31 were taking low-dose folic acid supplementation in pregnancy.

gestational age at randomization and gestational age at biomarker sampling. Time from randomization to biomarker sampling was calculated to account for time that participants were exposed to either the high- or low-dose study treatment. FA intake from supplements at the time of biomarker sampling was determined through combined assessment of FA supplementation at randomization, reported changes in medication use during the trial, and FA received through the study treatment. Dietary folate intake was assessed at the time of blood draw (24–26 weeks of gestation) through completion of the semiquantitative NutritionQuest Block Folic Acid/Dietary Folate Equivalent (DFE) Screener (www.nutritionquest.com), a brief screening tool consisting of 21 questions that approximates current or usual folate intake based on participant recall of the frequency of consumption of prespecified folate food sources. The block DFE screener has been shown to correlate with RBC folate concentrations in pregnant and nonpregnant women and has been shown to perform well against longer FFQs (33, 34).

Biological sampling and processing

Nonfasting peripheral blood samples were collected by venipuncture for whole blood, plasma, and serum analysis. Whole blood was collected into serum separator tubes or vacutainers coated with EDTA. EDTA whole blood was sent immediately for hematocrit analysis to the local on-site hospital laboratory. Otherwise, EDTA whole-blood specimens were kept on ice and wrapped in foil to protect from light until they were aliquoted and frozen (within 60 min of collection) or centrifuged for plasma separation (within 15 min of collection). Whole blood for plasma was centrifuged at 1500 $\times$ g for 15 min at 4°C and plasma immediately aliquoted and stored at –80°C. Serum separator tubes were inverted slowly as per product specifications and allowed to clot at room temperature (∼22°C) for 60 min. Samples were centrifuged at 1500 $\times$ g for 10 min at 4°C, and serum was aliquoted and stored at –80°C. Samples were shipped in batches every 6 wk to Health Canada in Ottawa, Canada, and stored at –80°C until analysis.

Biomarkers

Markers of folate status were analyzed (**Figure 2**), including serum and RBC total folate as well as individual serum folate vitamers—UMFA, tetrahydrofolate (THF), 5,10-methenylTHF, 5-formylTHF, 5-methylTHF, and MeFox (pyrazino-s-triazine derivative of 4α-hydroxy-5-methylTHF, an oxidation product of 5-methylTHF). Three additional biomarkers related to folate-mediated 1-carbon metabolism were measured, including total homocysteine, total vitamin B-12, and vitamin B-6 [pyridoxal 5′-phosphate (PLP)]. In the context of the FACT study, we were also interested in exploring the relation between FA intake and 3 angiogenic markers implicated in the development of preeclampsia—placental growth factor (PlGF), soluble endoglin (sEng), and soluble fms-like tyrosine kinase 1 (sFlt-1).

Serum and RBC total folate, plasma homocysteine, and serum vitamin B-12 were measured according to the manufacturer's instructions (Siemens ADVIA Centaur XP) at the Nutrition Research Canadian Health Measures Survey Reference Laboratory. Policies, procedures, and processes within the laboratory are aligned with licensing and accreditation standards of clinical laboratories in Canada and specifically the Institute for Quality Management in Healthcare, and assay performance targets (e.g., precision, bias, specificity, sensitivity) reflect their recommendations. All assays were verified prior to sample analysis using third-party quality control materials (BioRad). RBC total folate concentrations that fell above the highest calibrator were assigned the value of the highest calibrator for analyses.

LC-MS/MS was used to determine the contribution of individual folate vitamers to serum total folate. LC-MS/MS analyses were conducted using the US Centers for Disease Control and Prevention method adapted for manual wet chemistry as opposed to automated processing (35). Folate vitamers included THF, 5,10-methenylTHF and 5-formylTHF, 5-methylTHF, UMFA, and MeFox. In brief, serum samples were spiked with an internal standard mixture followed by addition of the extraction buffer (at pH 5) and incubated at 4°C for 30 min. Sample extraction and cleanup was performed by solid-phase extraction using sodium acetate elution buffer, and the final extracts were filtered through a 0.2-μm syringe filter prior to LC-MS/MS analysis. Folate peaks were separated by ultra performance liquid chromatography (UPLC) gradient mobile phase and detected by MS/MS in multiple reaction monitoring mode. Quantitation was performed using external calibration standards based on peak area ratio using internal standard. Quality control metrics for LC-MS/MS measurements are provided in **Supplemental Table 1**. Samples were analyzed using the Waters Xevo TQ-XS Mass Spectrometer coupled to the Waters Acquity UPLC system

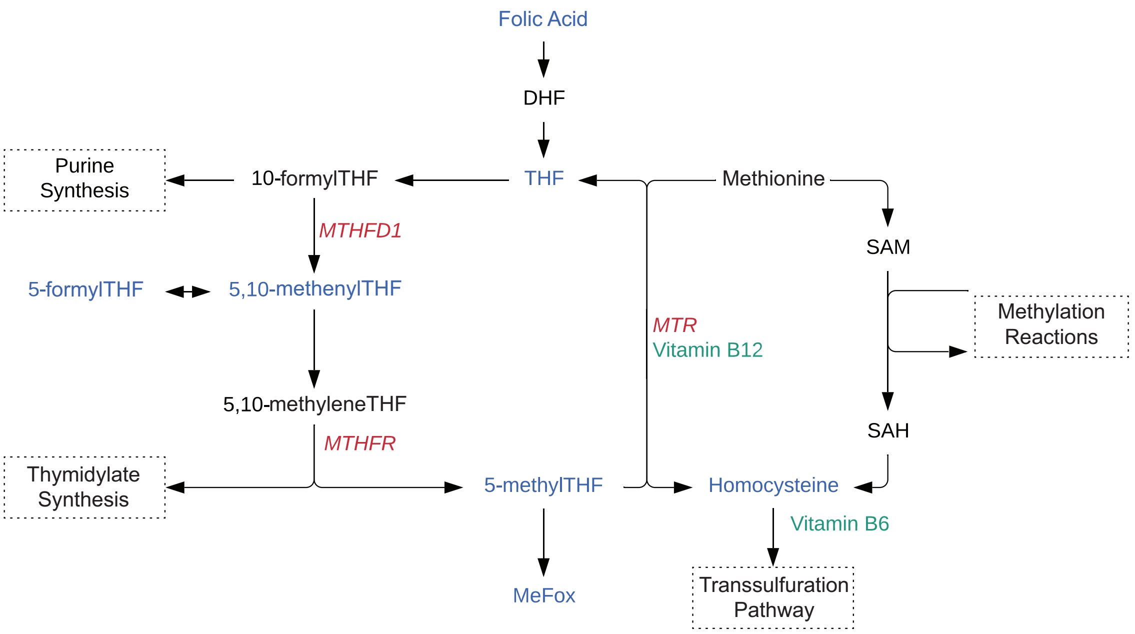

FIGURE 2 Folate-mediated 1-carbon metabolism. Analyzed metabolites (blue), vitamin cofactors (green), and enzyme genotypes (red). DHF, dihydrofolate; THF, tetrahydrofolate; MeFox, pyrazino-s-triazine derivative of 4α-hydroxy-5-methylTHF, a 5-methylTHF oxidation product; MTHFD1, methylenetetrahydrofolate dehydrogenase 1; MTHFR, methylenetetrahydrofolate reductase; MTR, methionine synthase; SAM, S-adenosylmethionine; SAH, S-adenosylhomocysteine.

(Waters Limited). Data collection and reduction was achieved using MassLynx software (V4.2). Folate vitamer data below the limit of detection (LOD) were assigned a value of LOD/2 for analyses.

The status indicator and coenzyme form of vitamin B-6, PLP, was measured in serum samples as the semicarbazone derivative by reversed-phase HPLC with fluorescence detection (36). Using low-, medium-, and high-control quality samples (ClinCheck 8873), the method performance was shown to have high precision (interassay CV of 4–8%) and accuracy (PLP concentrations of all levels in control ranges).

Angiogenic markers were analyzed using a customized Human Magnetic Luminex Assay according to the manufacturer's instructions (R&D Systems).

Genotyping

Three common single-nucleotide polymorphisms (SNPs) in folate-dependent enzymes were genotyped: methylenetetrahydrofolate reductase (*MTHFR*) 677C > T (rs1801133), methylenetetrahydrofolate dehydrogenase 1 (*MTHFD1*) 1958G > A (rs2236225), and methionine synthase (*MTR*) 2756A > G (rs1805087). DNA was isolated from whole blood using the Qiagen DNA Mini Kit. SNP genotyping was performed by qPCR using commercial TaqMan SNP Genotyping Assays (ABI-Life Technologies) with TaqPath ProAmp Master Mix (ABI-Life Technologies), according to the manufacturer's recommendations.

Ethics

FACT was approved by the Ottawa Health Science Network research ethics board (2009107–01H) and at all participating sites. FACT was registered at Current Controlled Trials (ISRCTN23781770). Both FACT and the biomarkers ancillary study were registered at clinicaltrials.gov as NCT01355159 and NCT03981029, respectively. The biomarkers ancillary study received separate approval from the research ethics boards associated with the Ottawa Health Science Network (2011649–01H), the Queen's University Health Sciences & Affiliated Teaching Hospitals (OBGY-226–12), the Horizon Health Network (2012–1736), and the University of British Columbia (H15–0,2812). Research ethics approval was also granted by the Health Canada and Public Health Agency of Canada Research Ethics Board (2011–0011).

Statistical analyses

All statistical analyses were conducted using SAS software, version 9.4 (SAS Institute) (37). The analyses are designed to characterize the study sample and compare high- and low-dose FA supplementation groups on demographic, clinical, and biomarker folate status and folate-mediated 1-carbon metabolism. Categorical variables are described as frequencies and percentages. Continuous variables are described with medians, IQRs, and minimum and maximum values, with the exception of the mean relative contribution of folate vitamers to serum total folate, which is described with mean and SD. In accordance with ethical protocols, all cell sizes <5 are suppressed to protect privacy. Differences and associations were considered significant when the P value was <0.05.

Differences in participant characteristics and measures of folate status and 1-carbon metabolism between the high- and low-dose supplementation groups were assessed. Chi-square tests and Fisher exact test (in the case of small cell sizes) were used to assess differences in categorical variables between groups. Wilcoxon rank-sum test was used to assess statistical differences in continuous variables between groups. Differences in mean relative contribution of folate vitamers to serum total folate were assessed by the Mann-Whitney rank-sum test.

Linear regression models were used to explore the unadjusted association between high- and low-dose FA on continuous measures of folate status and folate-mediated 1-carbon metabolism. Unadjusted associations were modeled by means of bivariable linear regression, resulting in unadjusted parameter estimates (slope coefficient, β) and corresponding SEs. The estimates represent a unit change in each biomarker when comparing high- to low-dose (reference group) FA. Multivariable linear regression models were used to adjust for the possible confounding effect of length of time on the study treatment.

Results

Study participants

A total of 251 women were randomly allocated into FACT at the 3 participating sites, of whom 51 participants consented to participate in the ancillary study. One participant had data for <50% of the biomarkers and so was excluded from the analysis. Of the 50 participants, 19 (38.0%) had been randomly allocated in FACT to the high-dose FA group and 31 (62.0%) to the low-dose FA group (Figure 1). The ancillary study participants were representative of FACT participants from the same study sites, and details are provided in **Supplemental Table 2**.

Baseline characteristics are summarized in **Table 1**. Participants taking high-dose and low-dose FA supplementation were similar for maternal age, prepregnancy BMI, level of education, employment status, and smoking history. The most common preexisting conditions reflecting preeclampsia risk factors included obesity (48.0%) and chronic hypertension (26.0%). Thirty-four percent of participants had a history of preeclampsia. A total of 48 (96.0%) women were taking FA supplements at FACT enrollment, and of these, 46 (95.8%) were taking 1.0 mg FA/d. A total of 14 women reported taking high-dose FA supplements (>1.1 mg/d) at any point before randomization. Genotype frequencies for *MTHFR* (rs1801133), *MTR* (rs1805087), and *MTHFD1* (rs2236225) were similar between the 2 groups (Table 1 and **Supplemental Table 3**).

Participant characteristics at the time of blood sampling are provided in **Table 2**. The median gestational age of participants at blood sampling was 25.0 wk (IQR: 24.4–26.1). Participants in the high-dose supplementation group spent more time on the study treatment compared with participants in the low-dose group (median: 12.0 wk, IQR: 11.0–13.3 compared with median: 10.9 wk, IQR: 10.0–12.0, respectively; $P = 0.009$). Estimated intake from FA supplements (regular supplementation + FACT treatment) was a median 5.0 mg/d (IQR: 5.0–5.0, range: 4.0–5.0) for the high-dose group and 1.0 mg/d (IQR: 1.0–1.0, range: 0.0–1.0) for the low-dose group. Total dietary folate equivalent DFE intake was similar among women in the high-dose group (median: 454.9 μg DFE, IQR: 375.8–716.9) and in the low-dose group (median: 556.6 μg DFE, IQR: 436.0–647.1). Intake of naturally occurring food folate and folic acid from fortified/enriched food sources did not differ between the study groups.

Folate status and markers of folate-mediated 1-carbon metabolism

The median concentration of RBC total folate in the study sample was 2701 nmol/L (IQR: 2243–3032) and did not differ between high-dose and low-dose supplementation arms (**Table 3**). All participants had RBC folate concentrations that exceeded the WHO 906-nmol/L threshold for optimal protection against NTDs. It should be noted that 35.4% of the RBC total folate values fell above the upper limit of our assay. The prevalence of off-curve high RBC total folate concentrations did not differ between the study groups (33.3% high dose compared with 36.7% low dose). RBC total folate did not differ between the groups when the analysis was restricted to values below the highest calibrator (data not shown).

In contrast, serum total folate concentrations were significantly higher in participants on high-dose FA. Serum total folate as measured by the protein binding assay was a median of 105.0 nmol/L (IQR: 57.0–183.0) in participants in the high-dose FA group compared with a median of 67.1 nmol/L (IQR: 58.5–89.0) in participants in the low-dose FA group ($P = 0.035$). Serum total folate as measured by LC-MS/MS was a median of 148.4 nmol/L (IQR: 110.4–181.2) in participants in the high-dose FA group compared with a median of 122.8 nmol/L (IQR: 99.5–136.0) in participants in the low-dose FA group ($P = 0.007$). Data from both assays are presented here to facilitate comparisons with other studies.

Serum 5-methylTHF concentrations were higher in the high-dose FA supplementation group (median: 126.6 nmol/L, IQR: 98.8–158.6) than the low-dose supplementation group (median: 108.6 nmol/L, IQR: 96.4–123.2), $P = 0.027$. Similarly, UMFA concentrations were higher among participants receiving high-dose supplementation (median: 4.6 nmol/L, IQR: 2.5–33.8) compared with those receiving low-dose supplementation (median: 1.9 nmol/L, IQR: 0.9–4.1), $P = 0.008$. A greater proportion of women in the high-dose group had UMFA concentrations >1.35 nmol/L (94.7 compared with 64.7%; $P = 0.018$), a cutoff that represents the 85th percentile for fasting UMFA concentrations in the prefortification Framingham Offspring Cohort (38). Concentrations of the nonmethyl folates and MeFox did not differ between the FA supplementation groups.

The mean relative contributions of folate vitamers to serum total folate differed between the 2 treatment groups. The mean relative contribution of UMFA was greater ($P = 0.025$) for women taking high-dose FA (mean ± SD: 13.4% ± 16.8%) compared with women taking low-dose FA (mean ± SD: 3.7% ± 4.9%; **Supplemental Figure 1**), whereas the mean relative contribution of 5-methylTHF was similar between the groups.

Plasma total homocysteine (median: 6.2 μmol/L, IQR: 5.1–7.8) did not differ between the groups, and no women had total homocysteine >11 μmol/L. Serum total vitamin B-12 (median: 215.0 pmol/L, IQR: 191.5–273.0) did not differ between the groups. The prevalence of serum vitamin B-12 <148 pmol/L was 6.3% and did not differ between the groups. Serum PLP (median: 40.9 nmol/L, IQR: 20.8–71.4) concentrations did not differ between the groups. The prevalence of serum PLP <20 nmol/L was 20.4% and did not differ between the groups.

TABLE 1 Baseline maternal and pregnancy characteristics of participants[1]

Characteristic	Overall, *n* (%)	High dose, *n* (%)	Low dose, n (%)	*P* value[2]
Total participants	50 (100.0)	19 (38.0)	31 (62.0)	
Hospital site				
Ottawa	22 (44.0)	7 (36.8)	15 (43.4)	0.618
Kingston	17 (34.0)	8 (42.1)	9 (29.0)	
Moncton	11 (22.0)	4 (21.1)	7 (22.6)	
Maternal age, y				
Median (IQR)	32 (30–35)	32 (30–37)	32 (28–34)	0.199
Min, max	20, 44	21, 44	20, 42	
Prepregnancy BMI (kg/m^2)				
Median (IQR)	33.6 (26.5–41.5)	33.0 (26.7–41.5)	36.3 (24.7–42.3)	0.648
Min, max	17.3, 53.5	20.2, 48.1	17.3, 53.5	
Gestational age at randomization, wk				
Median (IQR)	13.9 (12.4–15.3)	12.6 (11.9–14.0)	14.9 (13.4–15.7)	0.003
Min, max	8.1, 16.6	8.1, 16.0	10.1, 16.6	
Obstetrical risk factor				
Chronic hypertension	13 (26.00)	5 (26.32)	8 (25.81)	>0.999
Diabetes	<5	<5	<5	>0.999
History of preeclampsia	17 (34.00)	6 (31.58)	11 (35.48)	>0.999
Obesity (BMI ≥35)	24 (48.00)	7 (36.84)	17 (54.84)	0.216
Twin pregnancy	13 (26.00)	<5	<5	0.741
Caucasian race compared with other race	46 (92.0)	18 (94.7)	28 (90.3)	>0.999
Completed college or university	35 (70.0)	14 (73.7)	21 (67.7)	0.656
Full-time employment	40 (80.0)	17 (89.5)	23 (74.2)	0.282
History of smoking	27 (54.0)	11 (57.9)	16 (51.6)	0.665
Not smoking in current pregnancy	13 (48.1)	6 (54.5)	7 (43.7)	0.666
Quit prior randomization	<5	<5	<5	
Smoking at time of randomization	10 (37.0)	<5	7 (43.7)	
Alcohol intake during pregnancy				
No	47 (94.0)	18 (94.7)	29 (93.5)	>0.999
Quit during pregnancy	<5	<5	<5	
FA dose at randomization, mg/d				
Median (IQR)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	>0.999
Min, max	0.0, 1.0	0.0, 1.0	0.0, 1.0	
FA dose categories				
No supplementation	2 (4.0)	1 (5.3)	1 (3.2)	>0.999
Low dose (<1.1 mg/d)	48 (96.0)	18 (94.7)	30 (96.8)	
High dose (≥1.1 mg/d)	0	0	0	
History of high-dose FA supplementation (≥1.1 mg/d)	14 (30.0)	5 (26.3)	9 (32.3)	0.66
1.1–5.0 mg/d	9 (64.3)	4 (80.0)	5 (55.5)	0.580
≥5.0 mg/d	5 (33.3)	<5	<5	
How long prior to randomization did high-dose supplementation stop?[3]				
<1 wk	10 (71.4)	<5	7 (77.8)	0.550
≥1 wk	3 (21.4)	<5	<5	
MTHFR genotype frequency (C677T)				
CC	19 (38.0)	7 (36.8)	12 (38.7)	>0.999
CT	24 (48.0)	9 (47.4)	15 (48.4)	
TT	7 (14.0)	<5	<5	

[1]FA, folic acid; *MTHFR*, methylenetetrahydrofolate reductase. In accordance with privacy concerns, all cell sizes less than 5 are suppressed. Column statistics are provided. Data are presented as *n* (%) unless otherwise indicated.

[2]All comparisons were made by χ^2 test and Fisher exact test in the case of small cell sizes for categorical variables. Wilcoxon rank-sum tests were used for continuous variables.

[3]Missing data: high-dose group, $n = 2$; low-dose group, $n = 1$.

Adjustment for duration on FA treatment

Because the study groups differed in their duration on the FA treatment, a linear regression analysis was performed to adjust for that variable (**Table 4**). In unadjusted linear regression analyses, high-dose FA supplementation was associated with an increase in serum total folate when measured by both protein binding assay ($\beta = 109.25$, SE: 37.20) and LC-MS/MS ($\beta = 61.77$, SE: 18.86) (both $P \leq 0.005$). Similar results were found for 5-methylTHF

TABLE 2 Characteristics of participants at biomarker sampling (24–26 completed gestational weeks)

Characteristic	Overall, $n = 50$	High dose, $n = 19$	Low dose, $n = 31$	P-value[1]
Gestational age at biomarker sampling, wk				
Median (IQR)	25.0 (24.4–26.1)	24.6 (24.4–26.1)	25.3 (24.3–26.1)	0.557
Min, max	23.1, 28.4	24.0, 28.1	23.1, 28.4	
Time from randomization to biomarker sampling, wk				
Median (IQR)	11.4 (10.0–12.6)	12.00 (11.0–13.3)	10.86 (10.0–12.0)	0.009
Min, max	7.4, 16.3	10.1, 16.3	7.4, 16.3	
Total FA intake from supplements and study treatment,[2] mg/d				
Median (IQR)	1.0 (1.0–5.0)	5.0 (5.0–5.0)	1.0 (1.0–1.0)	<0.001
Min, max	0.8, 5.0	4.0, 5.0	0.8, 1.0	
Dietary folate intake (μg), median (IQR)				
Naturally occurring food folate (μg)	148.2 (99.0–189.8)	162.0 (99.0–227.8)	137.6 (96.2–189.8)	0.394
Fortified/enriched folic acid from food (μg)	226.6 (156.7–276.2)	176.3 (142.1–283.7)	243.7 (178.7–275.4)	0.236
Total food folate (μg)	346.7 (285.7–466.2)	318.3 (265.6–495.1)	376.6 (290.0–458.8)	0.538
Dietary folate equivalents from food (μg)	512.5 (393.4–655.8)	454.9 (375.8–716.9)	556.6 (436.0–647.1)	0.394

FA, folic acid.

[1]Differences assessed by Wilcoxon 2-sample test.

[2]Combined regular supplementation + Folic Acid Clinical Trial treatment.

(β = 24.61, SE: 8.50) and UMFA (β = 33.45, SE: 12.45). In multivariable linear regressions, after adjusting for length of time on the study treatment, these associations remained statistically significant. There were no associations between the high- and low-dose groups for RBC total folate, plasma total homocysteine, serum total vitamin B-12, serum PLP, THF, 5-formylTHF, 5,10-methenylTHF, or MeFox in unadjusted and adjusted analyses.

Angiogenic markers

PlGF, sEng, and sFlt-1 concentrations did not differ between the supplementation groups (**Supplemental Table 4**).

Discussion

Our findings demonstrate that women who initiate daily high-dose FA supplementation (4.0–5.1 mg) between 8 and 16 gestational weeks have higher serum total folate concentrations at 24–26 gestational weeks than women taking low-dose FA supplements (≤1.1 mg). The higher serum total folate concentrations in women taking high-dose FA were attributable to both higher UMFA and 5-methylTHF concentrations. There were no differences in nonmethylated folates, total homocysteine, and vitamins B-12 and B-6 between the groups. Genotype distributions of common SNPs in 1-carbon metabolism-related enzymes (MTR, MTHFR, MTHFD1) that could potentially influence the distribution of folate vitamers also did not differ

TABLE 3 Biomarkers of folate status and folate-mediated 1-carbon metabolism of participants

Characteristic	All, $n = 50$, median (IQR)	High dose, $n = 19$, median (IQR)	Low dose, $n = 31$, median (IQR)	P value[1]
RBC folate, nmol/L	2701 (2443–3032)[2]	2701 (2559–3017)[3]	2686 (2311–3047)[3]	0.966
Serum total folate,[4] nmol/L	74.4 (58.5–117.7)[3]	105.0 (57.0–183.0)	67.1 (58.5–89.0)[3]	0.035
Serum total folate,[5] nmol/L	128.6 (104.2–149.4)	148.4 (110.4–181.2)	122.8 (99.5–136.0)	0.007
5-MethylTHF, nmol/L	117.7 (96.6–129.6)	126.6 (98.8–158.6)	108.6 (96.4–123.2)	0.027
UMFA, nmol/L	2.8 (1.36–8.3)	4.6 (2.5–33.8)	1.9 (0.9–4.1)	0.008
THF, nmol/L	2.6 (0.8–9.9)	3.0 (1.0–12.6)	2.4 (0.6–8.5)	0.197
5-formylTHF, nmol/L	1.3 (0.8–1.6)	1.3 (0.8–1.6)	1.4 (0.6–1.6)	0.743
5,10-MethenylTHF, nmol/L	0.8 (0.4–1.1)	0.9 (0.5–1.6)	0.8 (0.2–1.0)	0.224
MeFox, nmol/L	5.4 (3.3–8.7)	6.9 (3.6–10.1)	5.0 (2.4–8.0)	0.326
Homocysteine, μmol/L	6.2 (5.1–7.8)[6]	6.0 (4.6–7.0)[2]	6.4 (5.6–7.8)[3]	0.274
Vitamin B-12, pmol/L	215.0 (191.5–273.0)[2]	217.5 (191.0–271.0)[3]	208.5 (192.0–277.0)[3]	>0.999
PLP, nmol/L	40.9 (20.8–71.4)[3]	44.3 (19.0–71.4)[3]	40.0 (21.2–80.9)	0.959

MeFox, pyrazino-s-triazine derivative of 4α-hydroxy-5-methylTHF, a 5-methylTHF oxidation product; PLP, pyridoxal 5′-phosphate; THF, tetrahydrofolate; UMFA, unmetabolized folic acid.

[1]Differences assessed by Wilcoxon 2-sample test, except for differences in prevalence of UMFA >1.35 nmol/L, which was assessed by Fisher exact 2-tailed test.

[2]Missing data: $n = 2$.

[3]Missing data: $n = 1$.

[4]Serum total folate measured by protein binding assay.

[5]Serum folate vitamers measured by LC-MS/MS. Serum total folate does not include MeFox.

[6]Missing data: $n = 3$.

TABLE 4 Bivariable and multivariable linear regression models to assess the unit change in each biomarker in the high-dose compared with low-dose folic acid supplementation groups

Outcome variables	Unadjusted model, $n = 50$		Adjusted model,[1] $n = 50$	
	β (SE)[2]	P value	β (SE)	P value
RBC folate, nmol/L	16.43 (136.68)	0.905	–73.76 (142.26)	0.607
Serum total folate,[3] nmol/L	109.25 (37.20)	0.005	111.49 (40.42)	0.008
Homocysteine, μmol/L	–0.60 (0.49)	0.227	–0.54 (0.52)	0.307
Vitamin B-12, pmol/L	–5.23 (18.54)	0.779	6.29 (19.38)	0.747
PLP, nmol/L	0.19 (16.35)	0.991	2.16 (17.63)	0.903
Total serum folate,[4] nmol/L	61.77 (18.86)	0.002	50.87 (19.97)	0.014
5-MethylTHF, nmol/L	24.61 (8.50)	0.006	21.21 (9.11)	0.024
UMFA, nmol/L	33.45 (12.45)	0.009	26.77 (13.23)	0.049
THF, nmol/L	2.36 (1.68)	0.165	1.97 (1.81)	0.283
5-formylTHF, nmol/L	–0.04 (0.19)	0.805	0.17 (0.19)	0.369
5,10-MethenylTHF, nmol/L	0.21 (0.16)	0.194	0.14 (0.17)	0.411
MeFox, nmol/L	1.19 (1.28)	0.356	0.61 (1.36)	0.658

MeFox, pyrazino-s-triazine derivative of 4α-hydroxy-5-methylTHF, a 5-methylTHF oxidation product; PLP, pyridoxal 5′-phosphate; THF, tetrahydrofolate; UMFA, unmetabolized folic acid.

[1]Adjusted for duration (weeks) on study treatment.

[2]β, the parameter estimate, corresponds to the unit change in the biomarker in high- compared with low-dose (reference) folic acid supplementation groups.

[3]Serum folate measured by protein binding assay.

[4]Total serum folate measured by LC-MS/MS.

between the groups. Our data suggest that higher FA intake in our sample of pregnant women increased circulating UMFA and 5-methylTHF but in the absence of differences to other markers of 1-carbon metabolism.

The median RBC total folate concentration of the women in our study was 2701 nmol/L, and all exceeded the WHO 906-nmol/L cutoff for optimal protection against NTDs (39). RBC total folate concentrations were very high in our sample regardless of participant FA dose, with ∼80% above the 97th percentile for RBC folate concentrations reported for Canadian women of reproductive age (2299 nmol/L; 95% CI: 2021, 2577) (7). The median RBC total folate concentration was also higher than that previously reported for other pregnant North American women (26, 40–43). For instance, the PREFORM (PREnatal FOlic acid exposuRe on DNA Methylation in the newborn infant) pregnancy cohort reported a mean RBC folate concentration of 2417 nmol/L (95% CI: 2362, 2472) among Canadian women at 12–16 gestational weeks (26). Furthermore, the median RBC folate concentration for both dose groups was higher than the 90th percentile RBC folate concentration (2253 nmol/L; 95% CI: 1956, 2603) in second-trimester pregnant women of the US NHANES 1999–2006 (41). Over a third of the women in our sample had RBC total folate concentrations higher than the upper limit of the assay, thereby limiting our ability to compare RBC total folate concentrations between the study groups. This, combined with differences in timing of folate measurements in pregnancy and in analytical assays used, also limited our ability to interpret our findings in the context of other studies beyond indicating the relatively high folate status of our study participants.

Women consuming 4–5.1 mg/d FA had higher serum total folate compared with those consuming ≤1.1 mg FA, driven by both higher 5-methylTHF and UMFA. Accumulation of UMFA occurs when FA intakes exceed metabolic capacity (44, 45). UMFA was detectable in all of the women; however, the concentrations were significantly higher in women taking high-dose FA than women taking low-dose FA. The UMFA concentrations observed in participants taking ≤1.1 mg FA were comparable to those observed in other Canadian studies (25, 26). However, UMFA in participants consuming 4.0–5.1 mg FA was 2 times higher than women at 12–16 gestational weeks taking 1 mg/d since early pregnancy (median plasma UMFA: 2.44 nmol/L, IQR: 2.01, 2.92) (26) and lactating women taking 1 mg/d from the first trimester to 8 wk postpartum (median serum UMFA: 2.21 nmol/L, 5th–95th percentiles: 0.31–87.4) (25). Serum UMFA concentrations in the high-dose group were also above the 90th percentile reported for the general US population (46). In addition, UMFA represented a higher proportion of serum total folate in the high-dose group, consistent with another study that found UMFA disproportionately increased when serum total folate exceeded 78.5 nmol/L (95% CI: 67.9, 89.1) in women consuming 1 mg/d FA from early pregnancy to 8 wk postpartum (25).

While the functional ramifications of circulating UMFA are largely unknown, it has been associated with, albeit inconsistently, reduced natural killer cell cytotoxicity (47). FA supplement use in pregnancy has also been associated with asthma and respiratory tract infections among exposed children (48–50). UMFA is hypothesized to inhibit folate-mediated metabolism through the inhibition of key enzymes (28–30), which could manifest as altered distributions of folate vitamers or functional folate deficiency indicated by higher homocysteine. For example, in vitro studies demonstrate that MTHFR can be inhibited by dihydrofolate, the product of the dihydrofolate reductase reduction of FA (29). Inhibition of MTHFR activity could replicate the effects of the *MTHFR* 677C > T SNP, which reduces enzyme activity and is associated with lower 5-methylTHF and higher nonmethylated folates in RBCs (51), as well as hyperhomocysteinemia (52). However, other than increased UMFA and 5-methylTHF, we did not observe differences in the distribution of nonmethylated serum folate vitamers or plasma total homocysteine. These observations do not necessarily reflect intracellular 1-carbon metabolism, but they do suggest limited effect on circulating biomarkers.

North American women of reproductive age with low NTD risk are recommended to consume 0.4–0.8 mg, whereas women at high risk for an NTD-affected pregnancy are recommended to consume 4–5 mg/d (11, 12). The recommended duration of supplementation throughout pregnancy and what constitutes "high risk" for an NTD vary considerably by jurisdiction (53), variability that is reflected in patient practices. In addition, the FA doses available in over-the-counter supplements will influence FA consumption. For example, in Canada, although a regular women's daily over-the-counter multivitamin generally contains 0.4–0.6 mg FA, prenatal over-the-counter multivitamins contain almost exclusively 1.0 mg (12). Furthermore, high-risk obstetrical patients, such as the women in our study who were at high risk for preeclampsia, are often prescribed FA doses >1 mg. Thus, Canadian women commonly consume FA at higher than the recommended 0.4 mg/d dose (54). In 2017, a national workshop convening stakeholders from academia, government, industry, and health care was held to identify and prioritize key challenges in harmonizing evidence-based guidelines and practices in FA use in pregnancy for low-NTD risk women (21). Among the 5 challenges identified was a need to mitigate the "more is better" phenomenon regarding vitamins among health care professionals and the public, where research and evidence has yet to investigate the potential harms and benefits of FA supplementation beyond the reduction of NTD risk. Although our findings do not indicate harm, they also do not demonstrate additional benefit given that all of the women exceeded the WHO cutoff for NTD risk reduction. Furthermore, the high UMFA concentrations suggest that the high-dose FA was supraphysiologic.

Our study is strengthened by our use of the FACT data set to provide detailed information on participant medical and obstetrical histories, as well as FA supplement use, including dosing and timing of initiation (31). Our evaluation of biomarkers of folate status and 1-carbon metabolism was comprehensive, capturing both RBC and serum folates, as well as individual folate vitamers, homocysteine, and vitamins B-6 and B-12. Also, dietary folate intake assessment established that differences in folate status biomarkers were due to the study supplement and not differences in background intake. The study also has its limitations. The study included a convenience sample nested in a larger trial, so selection bias limits our sample to individuals motivated to participate in both FACT and the current study. The lack of folate biomarkers at randomization and the single time-point evaluation of folate status in the mid-to-late second trimester also prevented an analysis of changes in folate status and 1-carbon metabolism over the course of pregnancy. In addition, fasting was not required prior to sample collection, and information on the timing of sample collection relative to ingestion of the daily FA supplement was not collected, likely contributing to variability in the analyzed biomarkers. The study population represented individuals at high risk for developing preeclampsia, with such clinical characteristics as high BMI, diabetes, and chronic hypertension, limiting the generalizability of the data to the general healthy population. Finally, the small sample size precluded meaningful evaluation of the association between markers of folate status and folate-related SNPs and the development of preeclampsia in the FACT cohort. Future work in this area would benefit from larger sample sizes to more robustly examine the impact of FA dose in pregnancy on markers of folate status and 1-carbon metabolism, while adjusting for possible confounding factors.

In conclusion, our findings provide insight into the effect of higher than recommended doses of FA supplements on folate status and biomarkers of 1-carbon metabolism in pregnancy. The folate status of all of the participating women was well above the WHO cutoff for NTD risk reduction, regardless of treatment group. UMFA was measurable in all women, and those consuming higher FA doses had markedly higher UMFA than those consuming lower doses. The findings from this study can inform evidence-based FA recommendations for pregnant women at higher risk for preeclampsia. Given that the FACT trial showed no effect of high-dose FA on preeclampsia risk (31), and we show here that there is no benefit from a metabolic perspective, high-dose FA is unwarranted for this clinical population.

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Data Availability

Data described in the manuscript, code book, and analytic code will be made available upon request in a deidentified form pending review and approval of a formal request to the FACT scientific review committee (mwalker@toh.ca; swwen@ohri.ca; rwhite@toh.ca).

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